GRAND CANYON NATIONAL PARK
WORLD HERITAGE SITE

REHABILITATE NORTH RIM INFRASTRUCTURE

VOLUME 4 – PIPELINES – ROARING SPRINGS CAVE TO ROARING SPRINGS PUMPHOUSE (RSPH)

VOLUME #	DESCRIPTION
VOLUME 0	GENERAL
VOLUME 1	PIPELINES - NORTH RIM
VOLUME 2	PIPELINES - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 3	PIPELINES - NORTH RIM TO SUPAI TUNNEL
VOLUME 4	PIPELINES - ROARING SPRINGS CAVE TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 5	ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 6	NORTH RIM WATER TREATMENT PLANT
VOLUME 7	NOT USED
VOLUME 8	NORTH RIM WASTEWATER COLLECTION SYSTEM
VOLUME 9	POWER DISTRIBUTION - NORTH RIM
VOLUME 10	POWER DISTRIBUTION - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 11	POWER DISTRIBUTION - ROARING SPRINGS PUMPHOUSE (RSPH) TO MANZANITA
VOLUME 12	POWER SYSTEM - COTTONWOOD CAMPGROUND

SHEET	SUB SHEET	SUB AREA	TITLE OF SHEET
1	040G001	GENERAL	SHEET INDEX
2	040C101	GENERAL	SITE PLAN
3	041X101	RSPH PIPELINE REHABILITATION	DEMOLITION PLAN
4	041C101	RSPH PIPELINE REHABILITATION	GESC PLAN
5	041C102	RSPH PIPELINE REHABILITATION	SHEET MAP
6	041C103	RSPH PIPELINE REHABILITATION	PLAN AND PROFILE STA 401+00 TO STA 407+71
7	041C401	RSPH PIPELINE REHABILITATION	ENLARGED SITE PLAN RS CAVE
8	041C501	RSPH PIPELINE REHABILITATION	CIVIL DETAILS WATERLINE REPLACEMENT
9	041C502	RSPH PIPELINE REHABILITATION	CIVIL DETAILS POWER AND FO INSTALLATION
10	041C901	RSPH PIPELINE REHABILITATION	PHOTO DETAILS
11	041E601	RSPH PIPELINE REHABILITATION	CAVE ONE LINE AND PANELBOARD SCHEDULE

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY Date Signed 5-1-2026 ARIZONA, U.S.A.

Nicholas Humphrey

Registered Professional Engineer CERTIFICATE NO. 41009 RYAN RHOADES Date Signed 5-1-26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD N. LYNCH TECH. REVIEW: M. GOUGH DATE: AUG 2026	SUB SHEET NO. 040 G001	TITLE OF SHEET GENERAL SHEET INDEX NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 1 OF 11

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GENERAL NOTES:

A. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0. SEE SHEET 002C105.

30 0 30 60

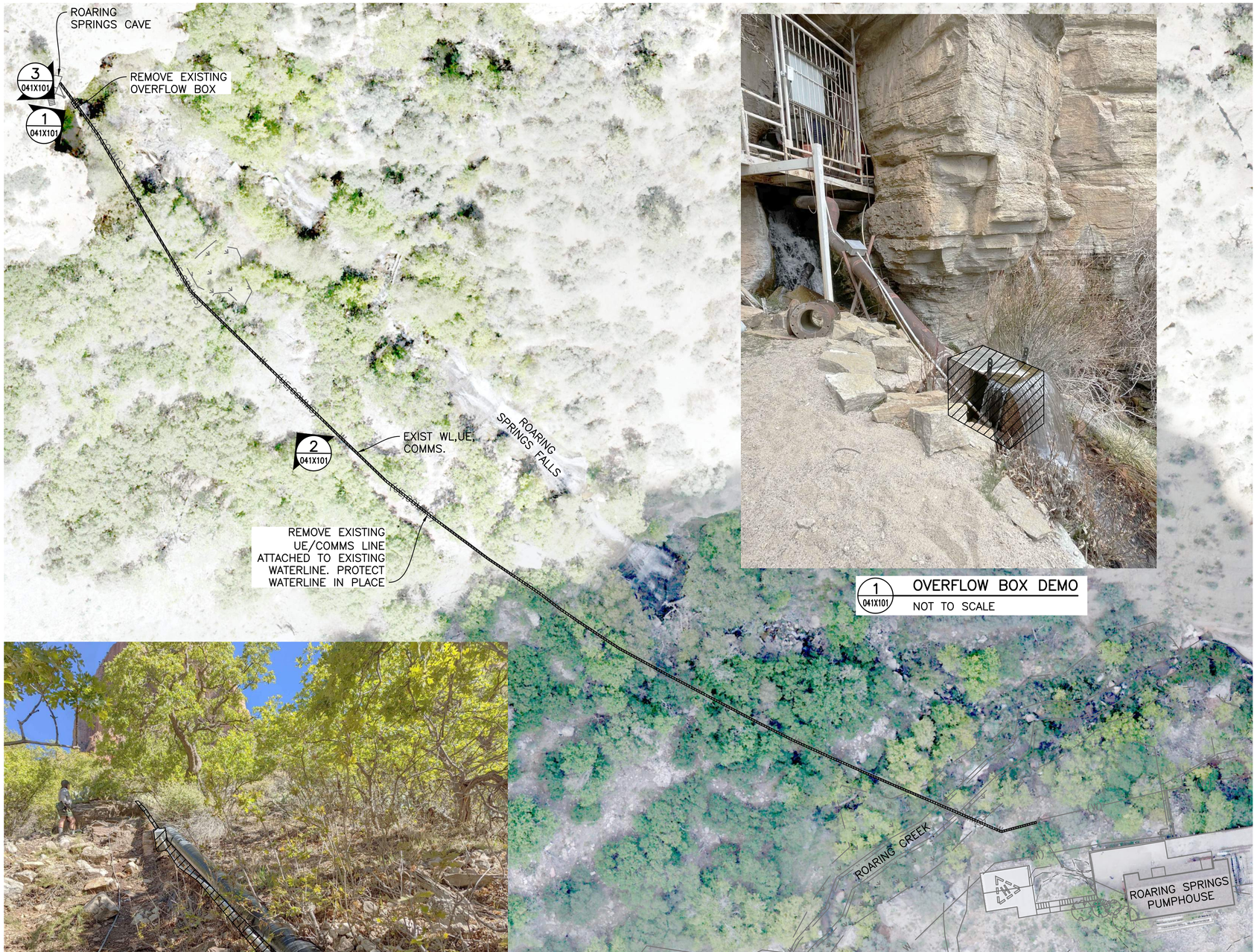
SCALE OF FEET
1" = 30'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-11-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 040 C101	TITLE OF SHEET GENERAL SITE PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 2 OF 11

4/16/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\103505545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\040-C101.dwg



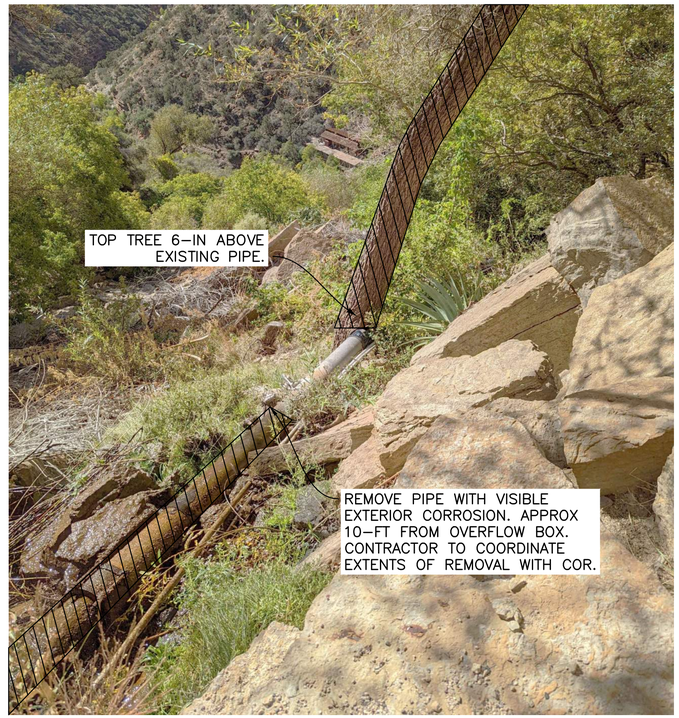
1 041X101 OVERFLOW BOX DEMO
NOT TO SCALE



2 041X101 ELECTRICAL AND COMMS DEMO
NOT TO SCALE

GENERAL NOTES:

A. CONTRACTOR TO PERFORM CCTV INSPECTION OF INTERIOR OF EXISTING PIPELINE PRIOR TO CONSTRUCTION AND IDENTIFY AREAS NEEDING REPLACEMENT. ANY REQUIRED REPAIRS TO BE COORDINATED WITH CONTRACTING OFFICER'S REPRESENTATIVE.



NOTE:

A. CONTRACTOR TO TOP TREE DOWNSTREAM OF EXISTING OVERFLOW BOX AND PROTECT EXISTING PIPE DURING REMOVAL.

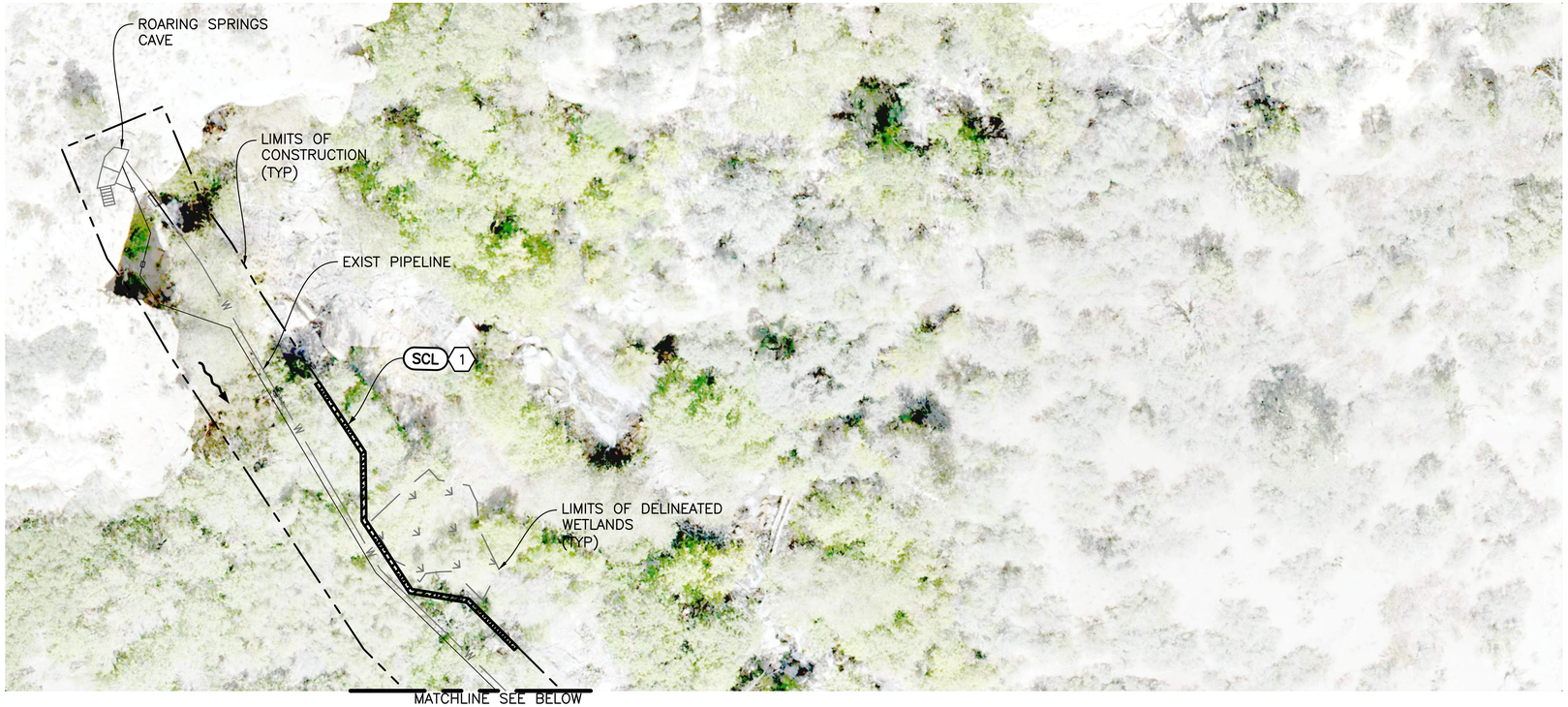
3 041X101 PIPE AND TREE REMOVAL
NOT TO SCALE

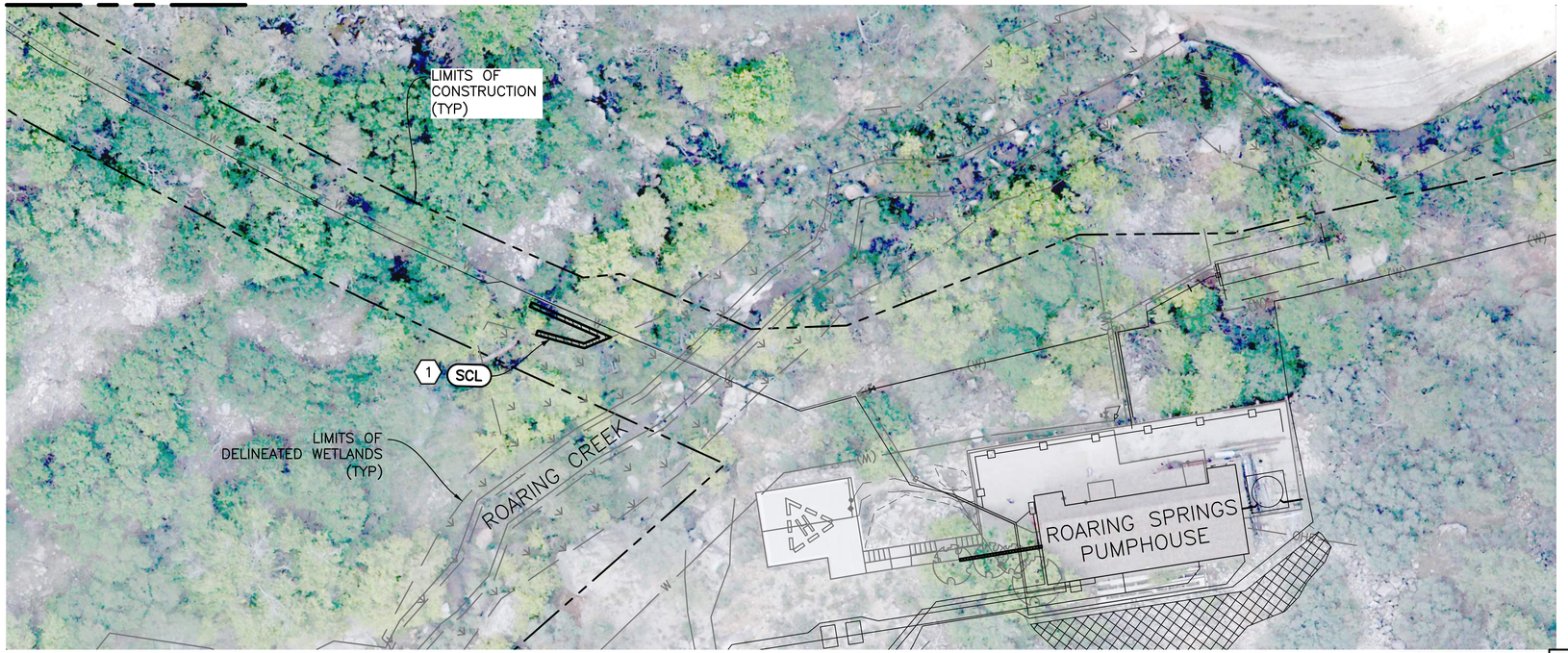
30 0 30 60
SCALE OF FEET
1" = 30'

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Nicholas Humphrey

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GENERAL NOTES:

A. FOR EROSION CONTROL NOTES AND DETAILS SEE VOLUME 0 SHEETS 002C509 – 002C512.

KEY NOTES: (X)

1. SEDIMENT CONTROL LOG
   1.1. QUANTITY = 145 LF (VOLUME 4 TOTAL)

LEGEND:

SCL — SEDIMENT CONTROL LOG, SEE 03/002C510

FLOW ARROW

20 0 20 40
SCALE OF FEET
1" = 20'

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Nicholas Humphrey

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30 0 30 60

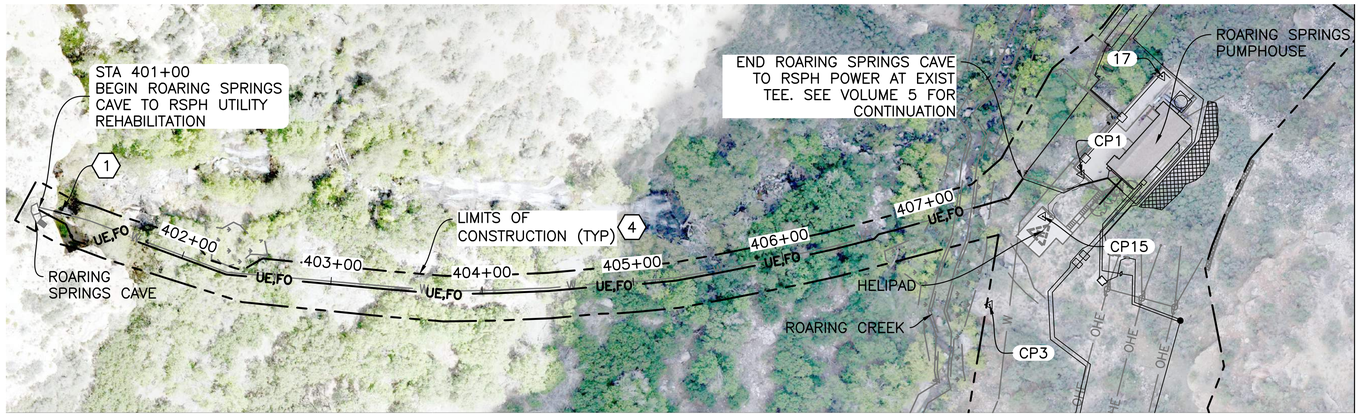
SCALE OF FEET
1" = 30'

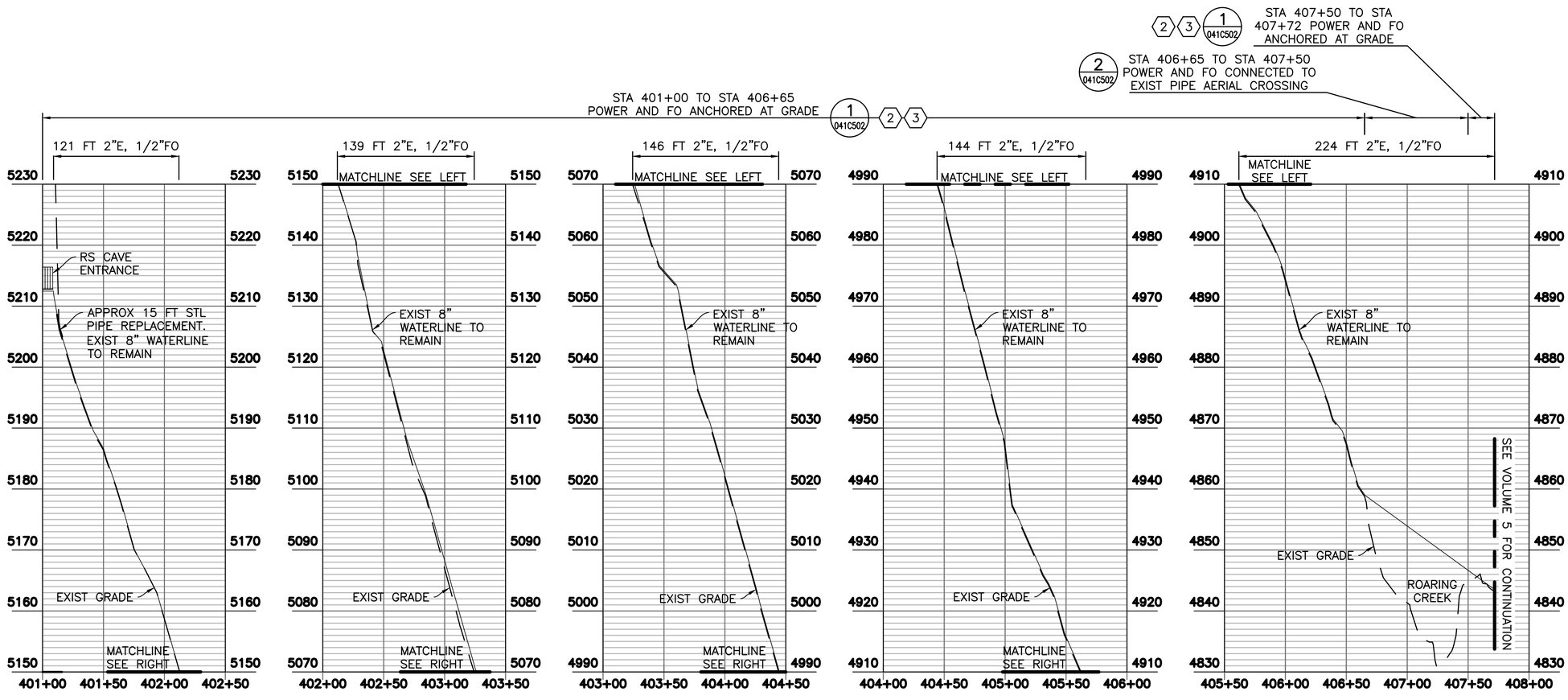
Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Not Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 041 C102	TITLE OF SHEET UTILITIES SHEET MAP NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 5 OF 11

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GENERAL NOTES:

A. LENGTHS OF UTILITIES SHOWN IN PROFILE REFLECTS CALCULATED SLOPE LENGTHS.

B. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0. SEE SHEET 002C105.

KEYNOTES: (X)

1. PIPE SEGMENT REPLACEMENT AT ABANDONED DIVERSION BOX. SEE DETAIL 1/041C501.
2. INSTALL JUNCTION BOXES AT A MAXIMUM SPACING OF 100 FT (SLOPE LENGTH). INSTALL OUTDOOR–WET RATED CONDUIT SUPPORT BUSHING WEDGES AT JUNCTION BOX CONDUIT ENTRANCES FOR POWER CABLE SUPPORT. PLACE FIRST CONDUIT SUPPORT BUSHING WEDGES NEAR THE RS CAVE ENTRANCE.
3. INSTALL ARMORED 24–STRAND SM FIBER W/ LC CONNECTORS ANCHORED AT GRADE.
4. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.

CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
401+00	406+65	12 FT	18 FT
406+65	407+50	15 FT	20 FT
407+50	407+72	SEE VOLUME 5	

50 0 50 100 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET
H: 1"=50' V: 1"=10'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 9-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. RACE (ADD) C. WITTENMEIER	041 C103	UTILITIES PLAN AND PROFILE STA 401+00 TO 407+71	113 182752
	TECH. REVIEW: GOUGH/LUSTER			PMIS/PKG NO. 318719
	DATE: AUG 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	SHEET 6 OF 11

8/12/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\041-C103.dwg

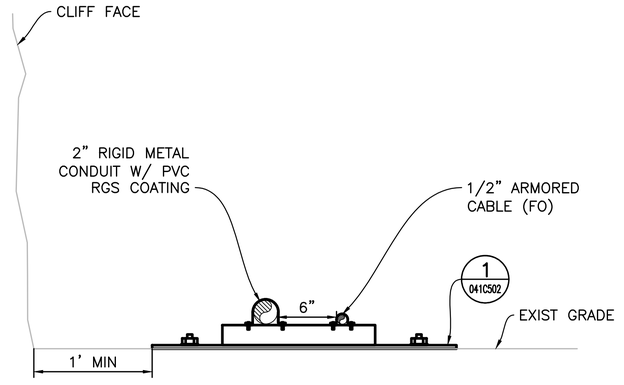


GENERAL NOTES:

A. ALL EQUIPMENT WILL NEED TO BE FLOWN INTO THE JOBSITE AREA.

KEYNOTES: (X)

1. TRANSFORMER SITING SHOWN ON PLANS IS APPROXIMATE, AND INTENDED TO BE ON LANDING OUTSIDE CAVE ENTRANCE. CONTRACTOR TO CONFIRM FINAL SITING AT CAVE AND UTILITY ROUTING WITH CONTRACTING OFFICER'S REPRESENTATIVE PRIOR TO CONSTRUCTION



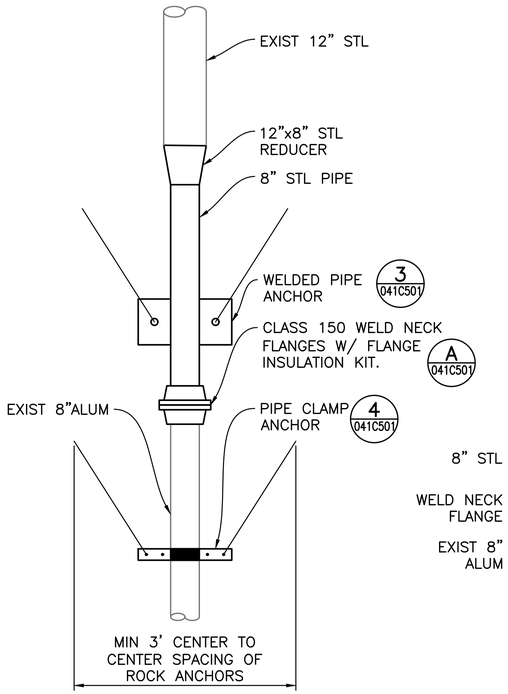
1 041C401 POWER AND FO INSTALLATION AT CAVE
NOT TO SCALE

5 0 5 10
SCALE OF FEET
1" = 5'

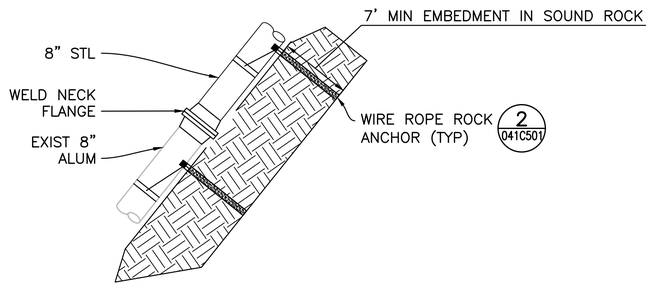
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Not Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 041 C401	TITLE OF SHEET UTILITIES ENLARGED SITE PLAN RS CAVE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 7 OF 11

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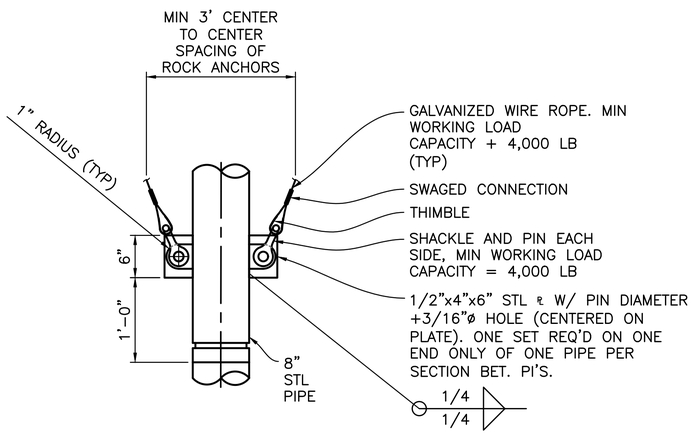
PLAN



PROFILE

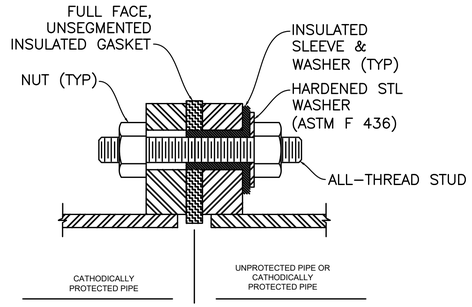
1 041C501 **RS CAVE PIPELINE REPLACEMENT AT OVERFLOW BOX**

NOT TO SCALE



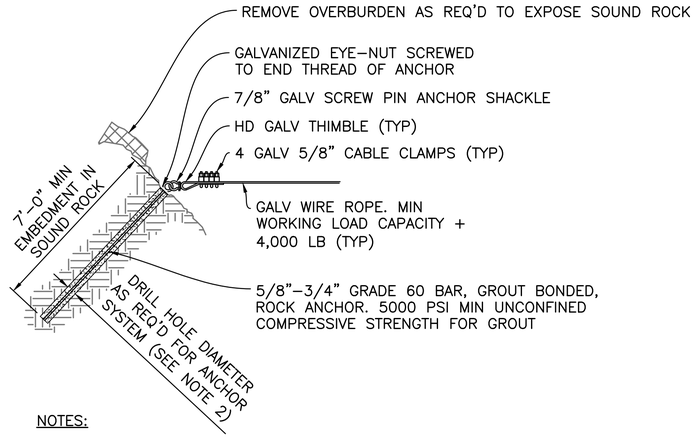
3 041C501 **WELDED PIPE ANCHOR PLAN**

NOT TO SCALE



A 041C501 **INSULATING FLANGE DETAIL**

NOT TO SCALE

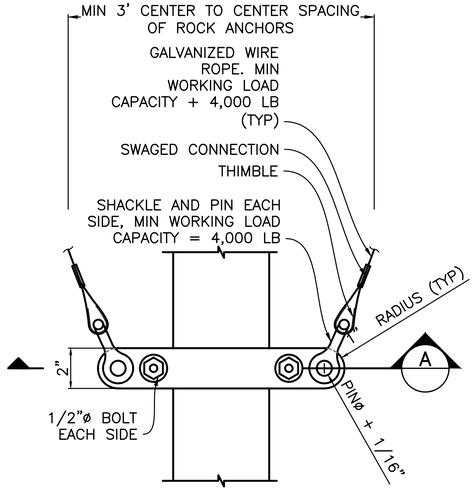


NOTES:

1. DRILL HOLE USING ROTARY PERCUSSION EQUIPMENT SUCH THAT DRILLED HOLE SIDE WALLS ARE ROUGH AND NOT SMOOTH, AS RECOMMENDED BY RESIN GROUT MANUFACTURER.
2. DRILLED HOLE DIAMETER SHALL BE 1-1/2" FOR 3/4" DIA. ROD.
3. GROUT BONDED ROCK ANCHORS TO BE DESIGNED AND SPECIFIED BY CONTRACTOR'S ENGINEER. ANCHORS SHALL HAVE A MIN COMBINED CAPACITY OF 8000 LB APPLIED IN ANY DIRECTION ON ANCHOR GROUP AND INDIVIDUAL ANCHOR CAPACITY OF 4000 LB SHEAR AND 4000 LB TENSION.
4. ALL STEEL HARDWARE SHALL BE GALVANIZED (G60).

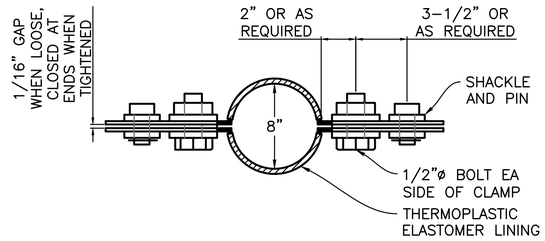
2 041C501 **WIRE ROPE ROCK ANCHOR**

NOT TO SCALE



4 041C501 **PIPE CLAMP ANCHOR**

NOT TO SCALE



A 041C501 **PIPE CLAMP SECTION**

NOT TO SCALE

NOTES:

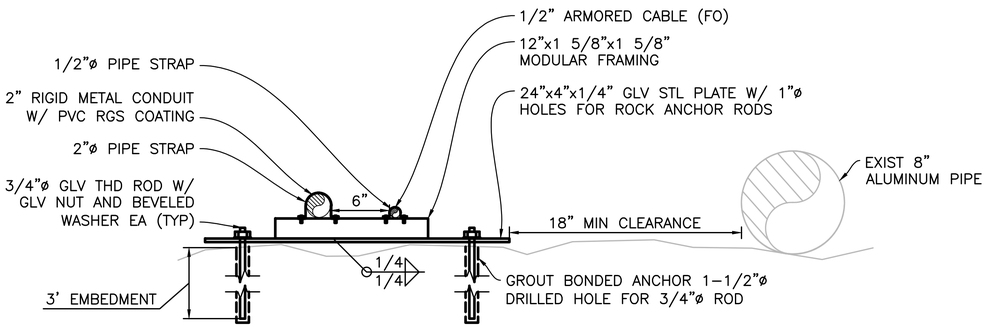
1. LOCATE ROCK ANCHOR POINTS AT EACH LOCATION PRIOR TO INSTALLING ANCHORS AND STEEL CABLES. CABLES SHALL BE PRECUT WITH THIMBLES AND SWAGED ENDS AND INSTALLED WITH HARDWARE SHOWN. ALL WIRE SHALL BE TAUT USING TURNBUCKLES AS SHOWN.
2. SLOPE OF WIRE ROPE SHALL BE NO LESS THAN 2V:1H.
3. ALL STEEL HARDWARE SHALL BE GALVANIZED.
4. GROUT BONDED ANCHORS SHALL HAVE A MIN COMBINED CAPACITY OF 8000 LB APPLIED IN ANY DIRECTION ON ANCHOR GROUP AND INDIVIDUAL ANCHOR CAPACITY OF 4000 LB SHEAR AND 4000 LB TENSION.
5. PROVIDE MOCK-UP OF PIPE CLAMP FOR APPROVAL BY CONTRACTING OFFICER PRIOR TO FABRICATION OF PIPE CLAMPS.
6. MOCK-UP SHALL BE LOAD TESTED BY CONTRACTOR FOR CAPACITY OF 8,000 LB FOR 24 HOURS. CONTRACTOR TO SUBMIT LOAD TEST PROCEDURE FOR APPROVAL PRIOR TO TESTING. CONTRACTING OFFICER SHALL BE INVITED TO WITNESS LOAD TEST. FINAL TEST REPORT SHALL BE SUBMITTED TO CONTRACTING OFFICER WITHIN 7 DAYS.





A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE DRAWN: C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 041 C501	TITLE OF SHEET UTILITIES CIVIL DETAILS WATERLINE REPLACEMENT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 8 OF 11

4/16/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\041-C501.dwg



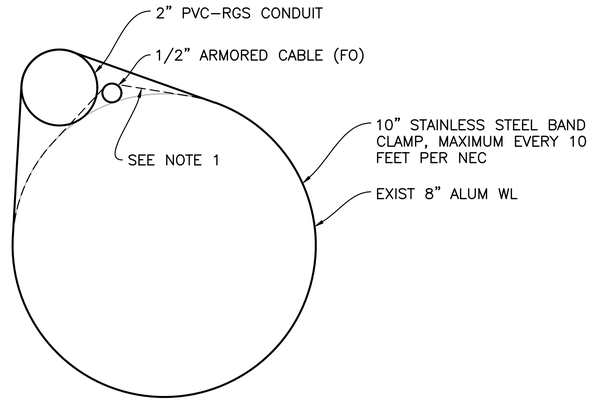
NOTES:

A. MAXIMUM CONDUIT SUPPORT SPACING 16' WHERE CONDUIT DOES NOT DEFLECT BETWEEN SUPPORTS. MAXIMUM SPACING 10' WHERE CONDUIT IS DEFLECTED BETWEEN SUPPORTS.

B. ALL HARDWARE TO BE GRADE 60 GLV STL.

1 / 041C502 POWER AND FIBER ANCHORED AT GRADE

NOT TO SCALE



NOTES:

1. PROVIDE BAND CLAMP FOR FIBER OPTIC CABLE EVERY 5 FEET ALONG PIPELINE, OFFSET FROM 10" ELECTRICAL BAND CLAMP.

2 / 041C502 POWER AND FO CONNECTED TO EXIST PIPE AERIAL CROSSING

NOT TO SCALE

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

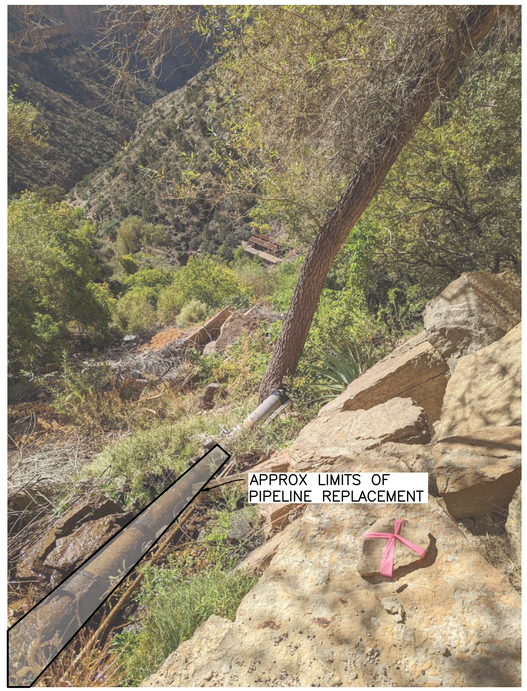
Nicholas Humphrey

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1 041C901 STA 401+10 PIPELINE REPLACEMENT
NOT TO SCALE



2 041C901 STA 401+25 PIPELINE REPLACEMENT EXTENTS
NOT TO SCALE

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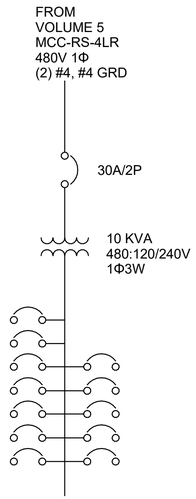
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

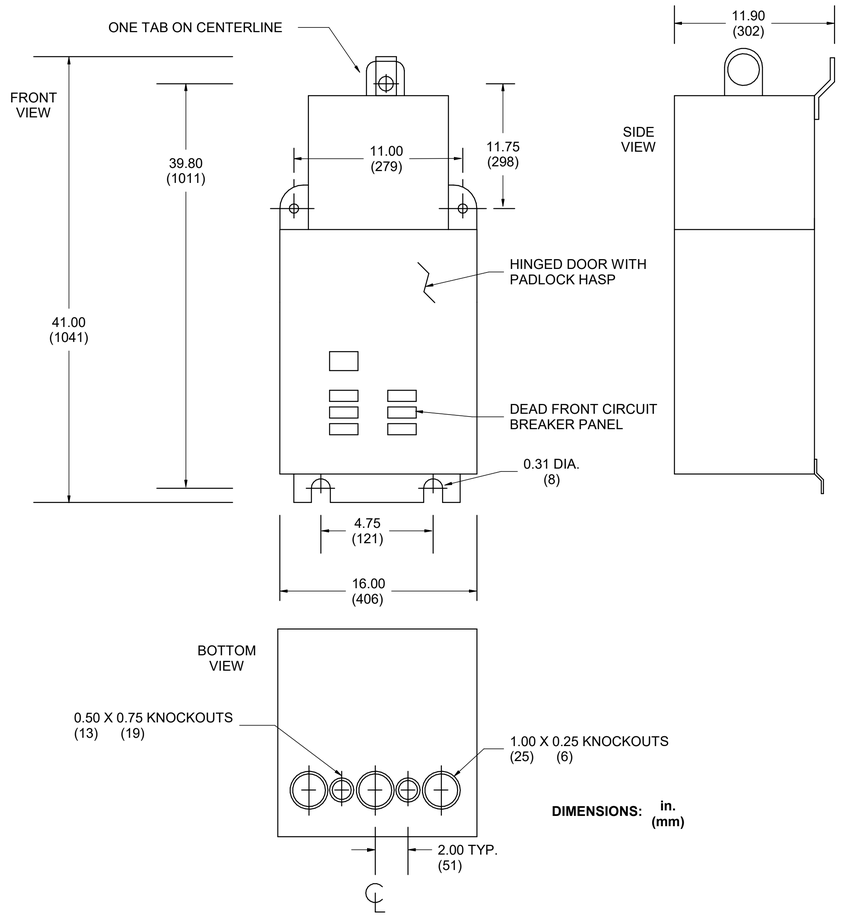
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE DRAWN: C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 041 C901	TITLE OF SHEET UTILITIES PHOTO DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 10 OF 11

KEYNOTES: ⟨X⟩

1. UTILIZE UP TO 3 BRANCH CIRCUITS TO FEED LOADS AT THE EXISTING CAVE CONTROL PANEL.



CAVE POWER ONE LINE



DIMENSIONS: in. (mm)

SINGLE PHASE ENCLOSURE AA

	PANELBOARD NO:	**RS-CAVE-MPZ**												
	VOLTAGE:	120/240		**BUS RATING (A):**		60A		**ENCLOSURE:**	NEMA 3R					
	PHASE:	1		**MAIN OC DEVICE (A/PHASE):**		60		**MOUNTING:**	SURFACE					
	WIRE:	3+GND		**INTERRUPTING RATING (KA):**		18		**LOCATION:**	CAVE					
	200% NEUTRAL:	NO		**SERVICE ENTRANCE LABEL:**		YES								
	CKT		**CONNECTED LOAD (VA)**			**OCP**			**OCP**		**CONNECTED LOAD (VA)**				**CKT**
	NO.	**DESCRIPTION**	**LTS**	**REC**	**POWER**	**AMPS**	**P**		**AMPS**	**P**	**LTS**	**REC**	**POWER**	**DESCRIPTION**	**NO.**
1	1	CIRCUIT 1			1,440	15	1	A	60	2				MAIN CIRCUIT BREAKER	2
1	3	CIRCUIT 3			1,440	15	1	B							4
	5	SPARE				20	1	A	15	1			1,440	CIRCUIT 2	6
	7	SPARE				20	1	B	20	1				SPARE	8
	9	SPARE				20	1	A	20	1				SPARE	10
	11	SPARE				20	1	B	20	1				SPARE	12

LOAD SUMMARY									
	LTS	REC	POWER	SPARE	TOTAL				PHASE BALANCE	
CONNECTED LOAD (KVA)	0.0	0.0	4.3	---	4.3		240	LINE-TO-LINE VOLTS	PHASE A (KVA)	2.9
DEMAND FACTOR	1.25	NEC	1.00	20%	---		18	CONNECTED AMPS	PHASE B (KVA)	1.4
DESIGN LOAD (KVA)	0.0	0.0	4.3	0.9	5.2		22	DESIGN AMPS		

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REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 58037 — TRAVIS T. MOORE — Date Signed 1/16/2026 — ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	J. HUCKENPAHLER	041 E601	RSPH PIPELINE REHABILITATION CAVE ONE LINE AND PANELBOARD SCHEDULE	113 182752
	S. BILD			PMIS/PKG NO. 318719
	TECH. REVIEW: P. DAVILA		NORTH RIM - GRAND CANYON NATIONAL PARK	SHEET 11 OF 11
	DATE: MAY 2026			

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